

1. SSID = 30 Munroe St.

SSID = linksys12

```
SSID="linksys"
, BI=100, SSID="30 Munroe St"
, BI=100, SSID="30 Munroe St"
, BI=100, SSID="30 Munroe St"
, BI=100, SSID="30 Munroe St"
, BI=100, SSID="30 Munroe St"
, BI=100, SSID="30 Munroe St"
, BI=100, SSID="30 Munroe St"
BI=100, SSID="30 Munroe St"
ags=....R..TC
ags=...P...TC
BI=100, SSID="linksys12"
, BI=100, SSID="30 Munroe St"
```

2. 30 Munroe St - 0.1024 sec

```
IEEE 802.11 Wireless Management
  Fixed parameters (12 bytes)
    Timestamp: 174319412160
    Beacon Interval: 0.102400 [Seconds]
```

Linksys also 0.1024 sec

```
IEEE 802.11 Wireless Management
  Fixed parameters (12 bytes)
    Timestamp: 9534921933578
    Beacon Interval: 0.102400 [Seconds]
```

3. Source address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)

4. Destination address: Broadcast (ff:ff:ff:ff:ff:ff)

5. BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)

6. Tag: Extended Supported Rates 6(B), 9, 12(B), 18, 24(B), 36, 48, 54, [Mbit/sec]

7.a. Receiver address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
.....0. = LG bit: Globally unique address
.....0. = IG bit: Individual address (unicast)
Transmitter address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
.....0. = LG bit: Globally unique address
.....0. = IG bit: Individual address (unicast)
Destination address: CiscoLinksys_f4:eb:a8 (00:16:b6:f4:eb:a8)

7.b. The wireless host > 00:13:02:d1:b6:4f

7c. First-Hop Router > 00:16:b6:e8:1b:30

7.d. Source IP: 192.168.1.109

7.e. Destination IP: 128.119.245.12

```
192.168.1.109      128.119.245.12
```

7.f. Some other network-attached device, destination IP is not the wireless host, and the evidence is shown in the trace, the MAC destination is the router, not the AP or host.

```
8. Receiver address: 91:2a:b0:49:b6:4f (91:2a:b0:49:b6:4f)
    .... ..0. .... = LG bit: Globally unique address
    .... ..1. .... = IG bit: Group address (multicast)
Transmitter address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
    .... ..0. .... = LG bit: Globally unique address
    .... ..0. .... = IG bit: Individual address (unicast)
Destination address: 91:2a:b0:49:b6:4f (91:2a:b0:49:b6:4f)
```

8.b. Host: 00:13:02:d1:b6:4f

8.c. AP: 00:16:b6:f7:1d:51

8.d. First-hop: AP delivers directly to host

8.e. Sender MAC - 00:16:b6:f7:1d:51, IP Sender: 128.119.245.12

```
9. No.      Time           Source           Destination      Protocol Length Info
    1733  49.583615    192.168.1.109    192.168.1.1      DHCP             390      DHCP Release - Transaction ID 0xea5a526
```

```
1751  49.653218    Intel_d1:b6:4f    CiscoLinksys_f5:ba:bb    802.11107
Association Request, SN=1607, FN=0, Flags=....R...C, SSID="linksys_SES_24086"
```

```
10. 1735      49.609617    Intel_d1:b6:4f    CiscoLinksys_f7:1d:51    802.1154
Deauthentication, SN=1605, FN=0, Flags=.....C
```

```
1740  49.638857    Intel_d1:b6:4f    CiscoLinksys_f5:ba:bb    802.1158
Authentication, SN=1606, FN=0, Flags=.....C
```

Two authentication attempts, both fail as there is no reply from AP.

11. Host wants open authentication

```
Fixed parameters (6 bytes)
Authentication Algorithm: Open System (0)
```

12. No, there is no reply from linksys_ses_24086 AP to host authentication requests.

13. AP accepts requests, unlike linksys_ses_24086, allows the host to proceed with reassociation.

```
2156 63.168087 Intel_d1:b6:4f CiscoLinksys_f7:1d:51 802.1158
Authentication, SN=1647, FN=0, Flags=.....C
2158 63.169071 CiscoLinksys_f7:1d:51 Intel_d1:b6:4f 802.1158
Authentication, SN=3726, FN=0, Flags=.....C
```

```
14. Associate request: 2158 63.169071 CiscoLinksys_f7:1d:51
Intel_d1:b6:4f 802.1158 Authentication, SN=3726, FN=0, Flags=.....C
Associate response: 2164 63.170692 CiscoLinksys_f7:1d:51 Intel_d1:b6:4f
802.1158 Authentication, SN=3727, FN=0, Flags=.....C
```

15. Host and AP both support:

1.0, 2.0, 5.5, 11.0 (802.11b)

6.0, 9.0, 12.0, 18.0, 24.0, 36.0, 48.0, 54.0 Mbps (802.11g)

No tagged parameters